

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2018

Bachelor in Computer Applications Full Marks: 60
Course Title: Digital Logic Pass Marks: 24
Code No: CACS105 Time: 3 hours
Semester: I

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. [6 × 5 = 30]

11. Subtract: $675.6 - 456.4$ using both 10's and 9's complement. [5]

12. What is universal logic gate? Realize NAND and NOR as universal logic gates.

[1 + 2 + 2]

13. Simplify (using K-map) the given Boolean function F in both SOP and POS using don't care conditions D :

DCBADCB ■■■■■

DABCDDBCDBF ■■■■■■■■■ [2 + 3]

14. Define encoder: Draw logic diagram and truth table of octal - to - binary encoder. [1 + 4]

15. What is D flip-flop? Explain clocked RS flip-flop with its logic diagram and truth table.

[1 + 4]

16. Design MOD - 5 counter with state and timing diagram. [2 + 1 + 2]

17. Design a 4 - bit serial into parallel - out shift register with timing diagram. [3 + 2]

Group C

Attempt any TWO questions. [2 × 10 = 20]

18. Write difference between PLA and PAL. Design a PLA circuit with given functions.

$F_1(A, B, C) = \Sigma(2, 3, 5)$

$F_2(A, B, C) = \Sigma(0, 4, 5, 7)$. Design PLA program table also. [3 + 7]

19. Define D flip-flop. Design a Master-slave flip-flop by using JK flip-flop along with its circuit diagram and truth table. [2 + 8]

20. Write down the difference between asynchronous and synchronous counter. Design a 4 -

bit binary ripple counter along with its circuit, state and timing diagram. [3 + 7]